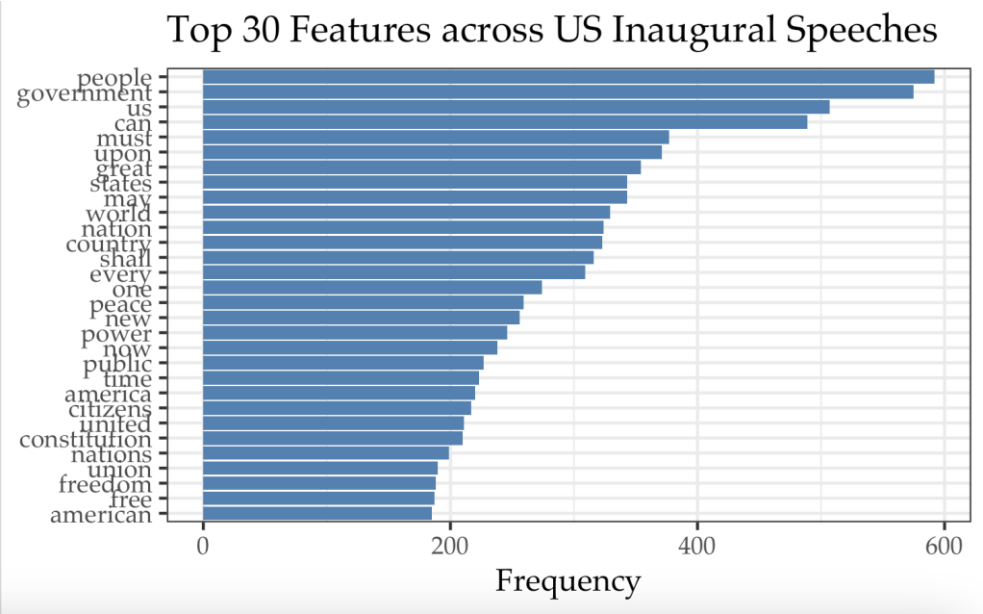
Tompkins – Assignment 4

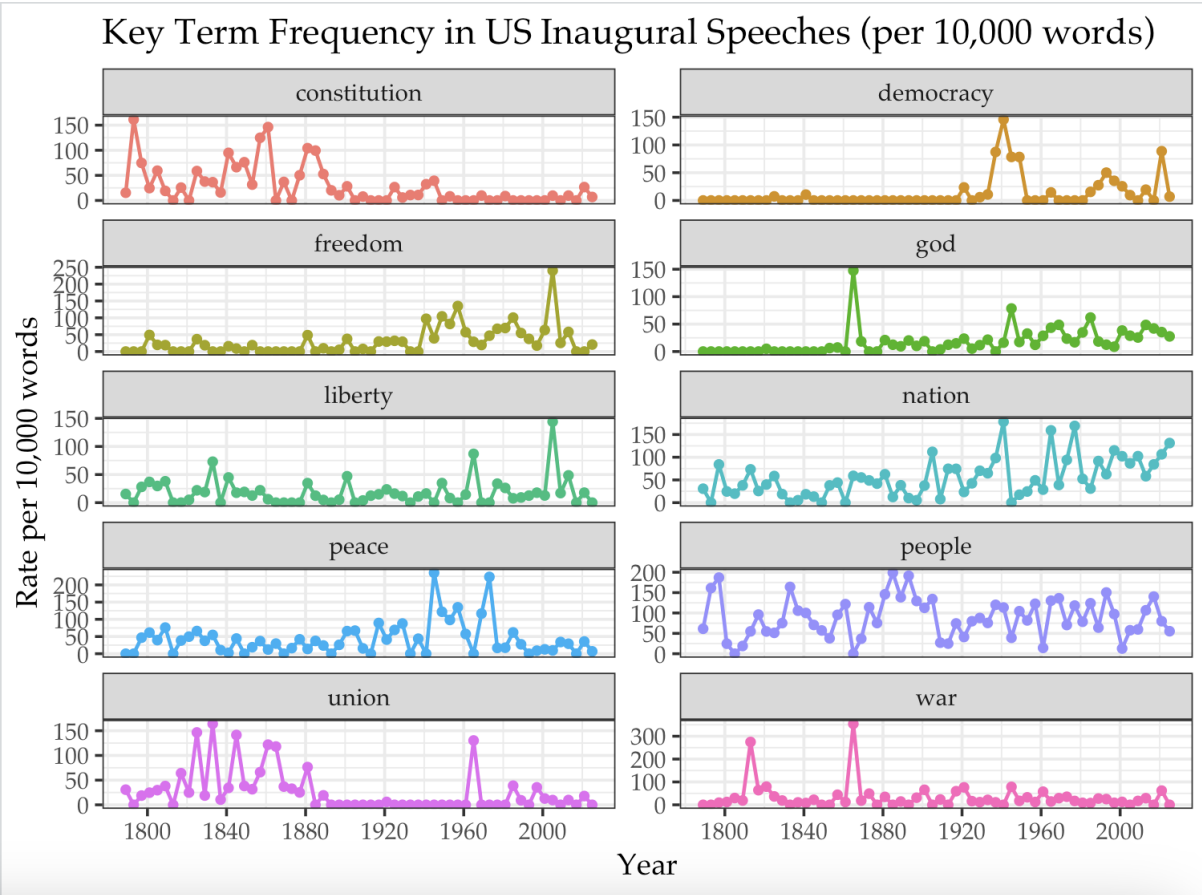
Use sample programs from Teams to analyze:

a. US presidential inaugural speeches i. Any similarities and differences over time and among presidents?

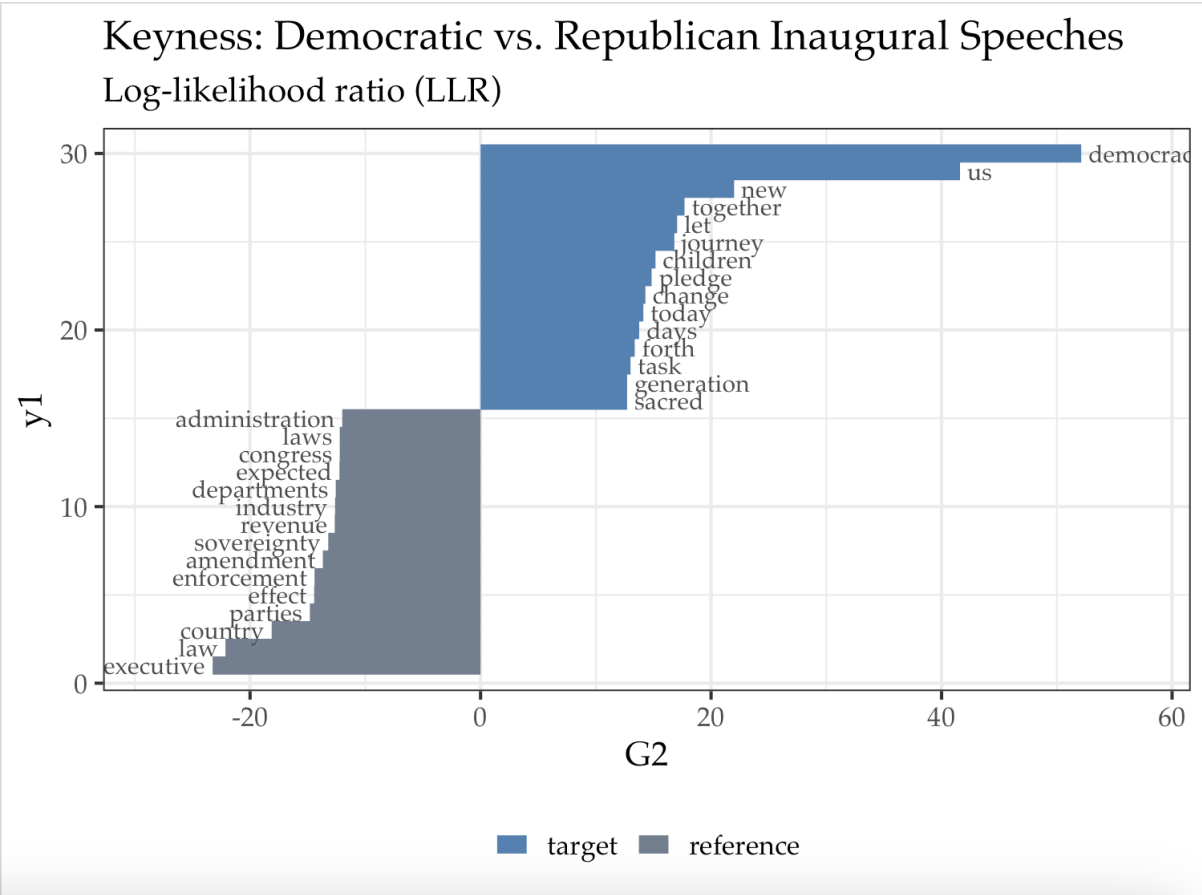
The first figure shows the top 30 words that feature the most across US inaugural speeches from 1789 to 2021. We see that human-centered words that evoke a sense of national community like “people”, “us”, “citizens” and “united” feature very frequently across these speeches, suggesting that a traditional aim of these speeches may be to unite the country after an election and explain how the president will serve this imagined national community.



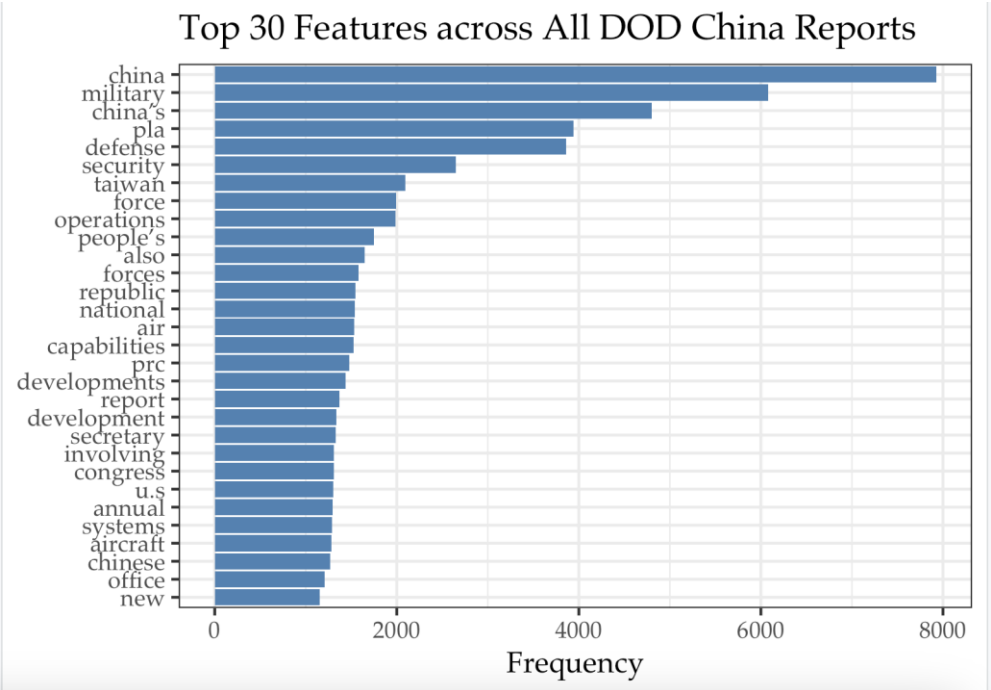
The next figure shows key term frequency for author-selected words including: “constitution”, “democracy”, “freedom”, “god”, “liberty”, “nation”, “peace”, “people”, “union” and “war”. Freedom, democracy and liberty are rarely mentioned in older speeches and are much more commonly featured in more recent speeches, whereas constitution features more heavily in earlier speeches.



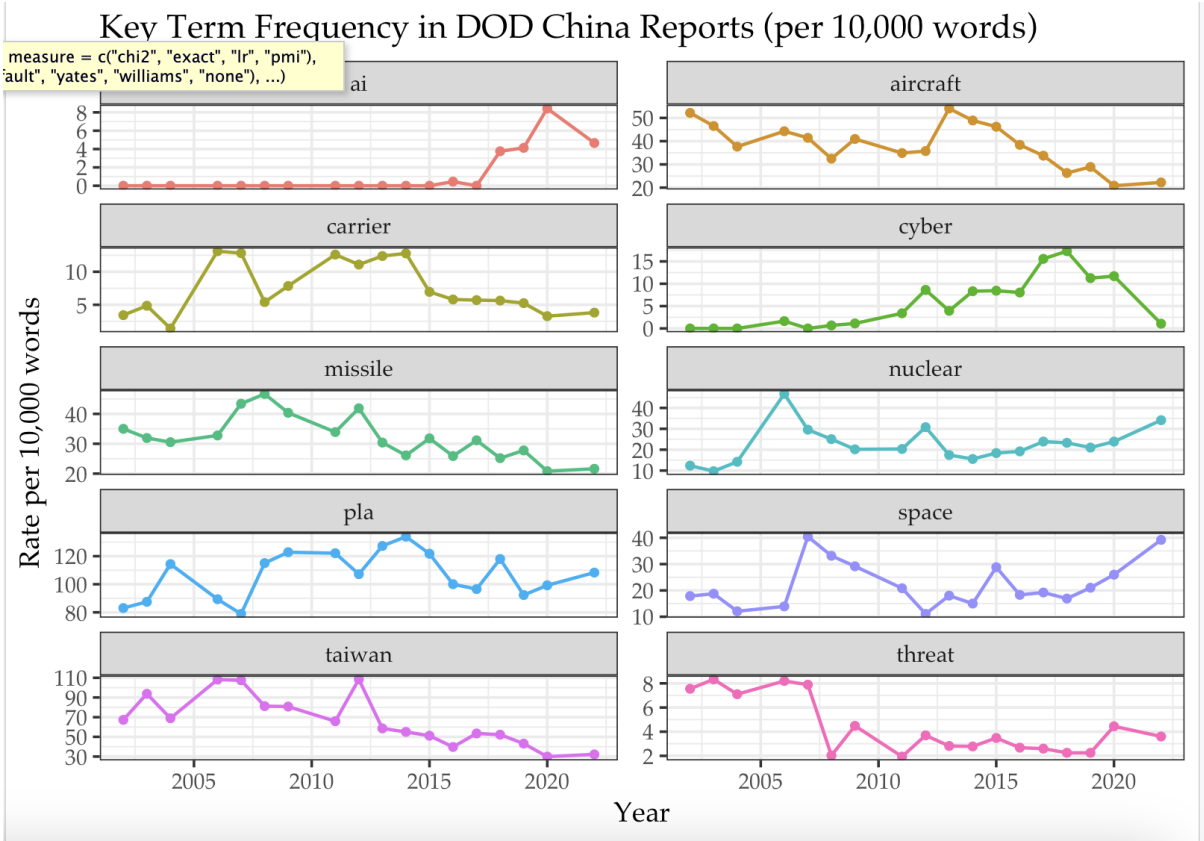
The final figure shows the difference in words used between Democratic and Republican speeches across time. We see that democratic speeches seem to focus more on community “us”, “together”, “generation”, while Republican speeches seem to focus on “harder” topics such as enforcement and laws.



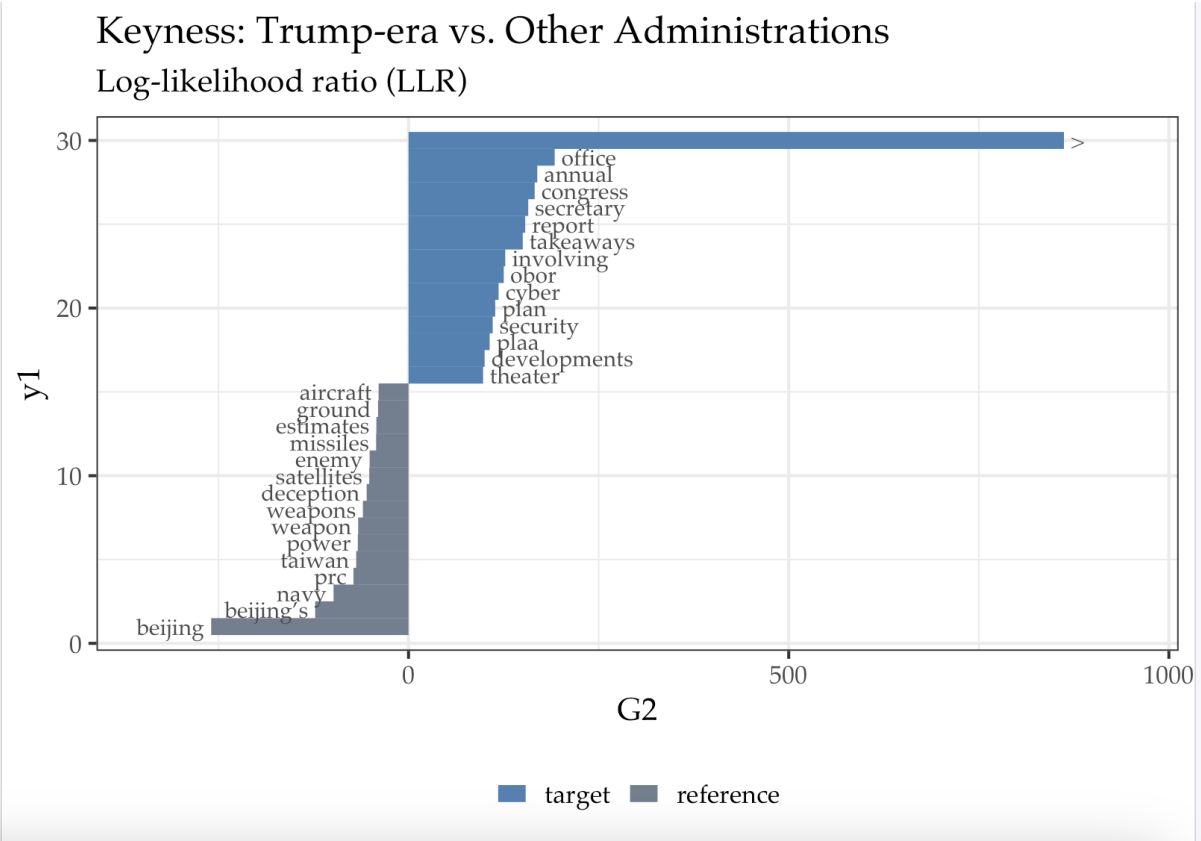
b. US Department of Defense reports (USDOD.zip in Teams Labs folder)



This trend analysis explores the top thirty most common words used across all DOD reports in my sample (2002 – 2022). Unsurprisingly "China" and "china's" were some of the most frequently used words, as were other related words like chinese and Taiwan.



This figure shows the frequency of key words over time. We see a rise in AI and cyber over time, and interestingly a decline in Taiwan and threat. This could potentially indicate a loss of interest in Taiwan over time.

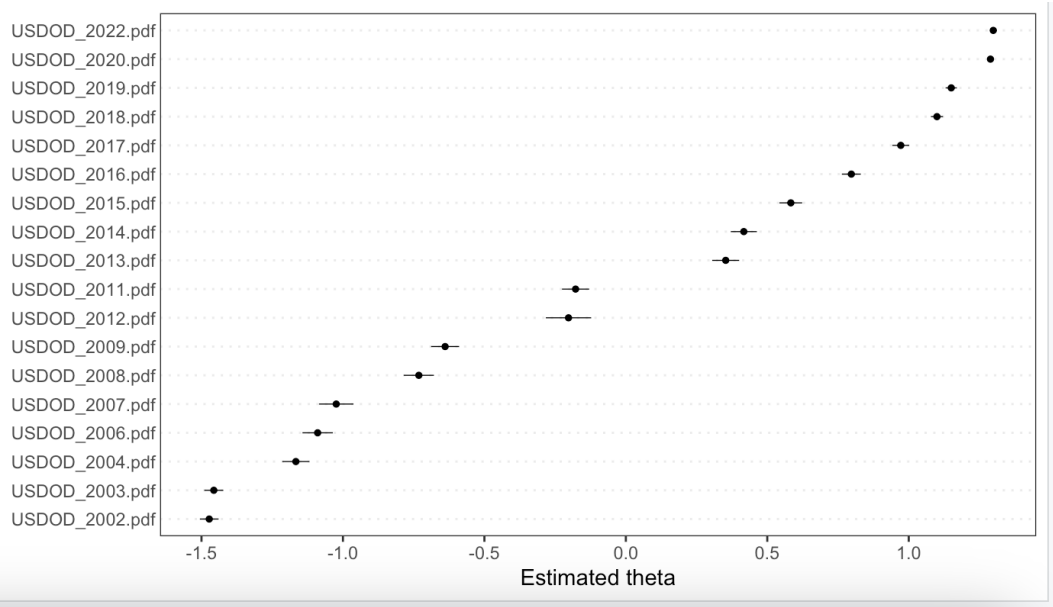


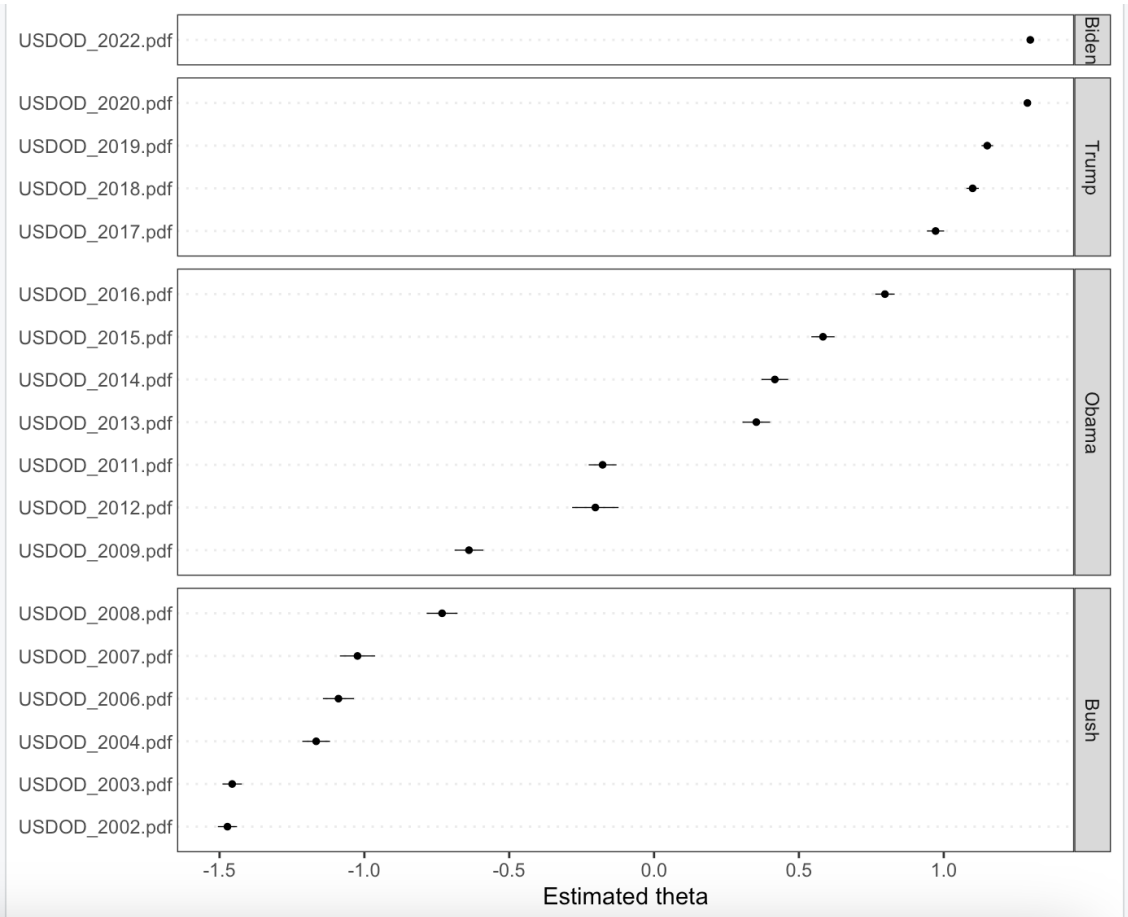
This figure whos the words that have been more frequently used in the Trump administration compared to previous administrations. We see an increased focus on cyber, and a decreased focus on physical military concepts like missiles and aircraft.

What is Wordfish?

Wordfish is a scaling tool that positions documents along a scale based on word frequencies. The scale was originally designed to position political documents on an ideological scale (i.e. left to right) without any human judgement. It does this by assessing which words are used more by one side than the other. Its main score is theta, which is the position score assigned to each document on the scale. It also estimates alpha (document length/word richness), beta (how strongly a word pulls towards one end of the scale) and psi (each word's baseline frequency).

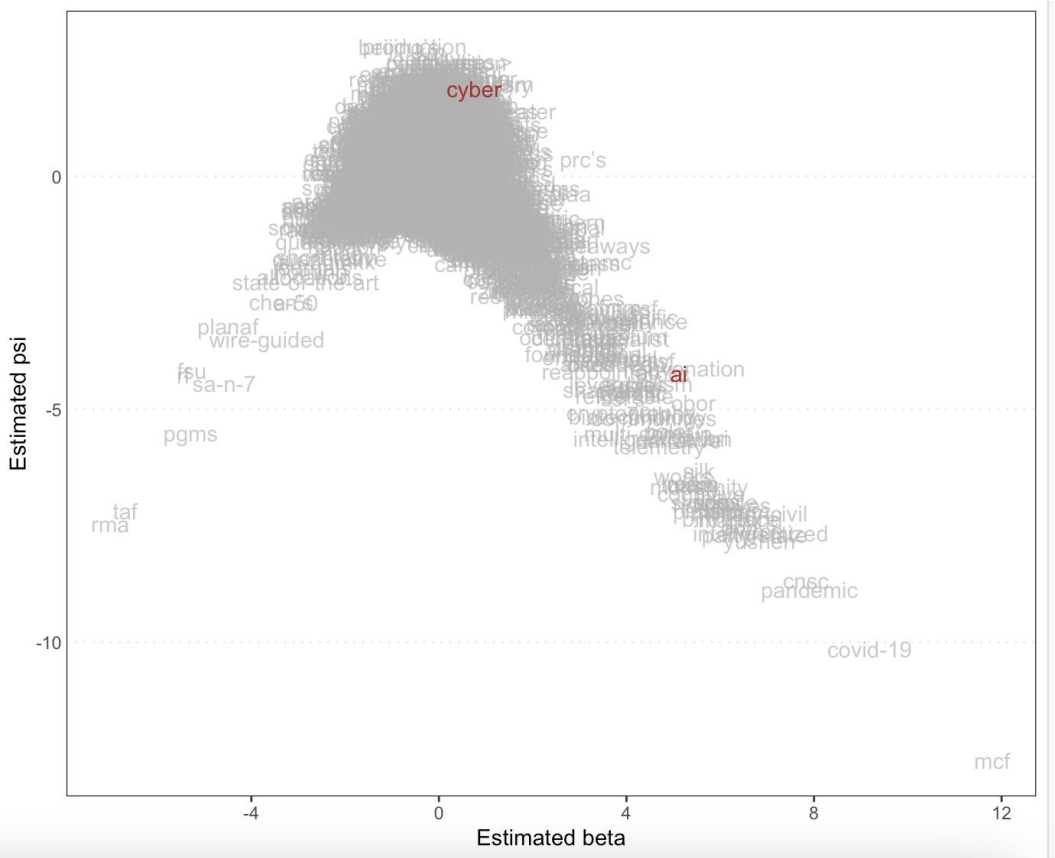
Run Wordfish on DOD reports – note I ran them on reports from 2002 to 2022



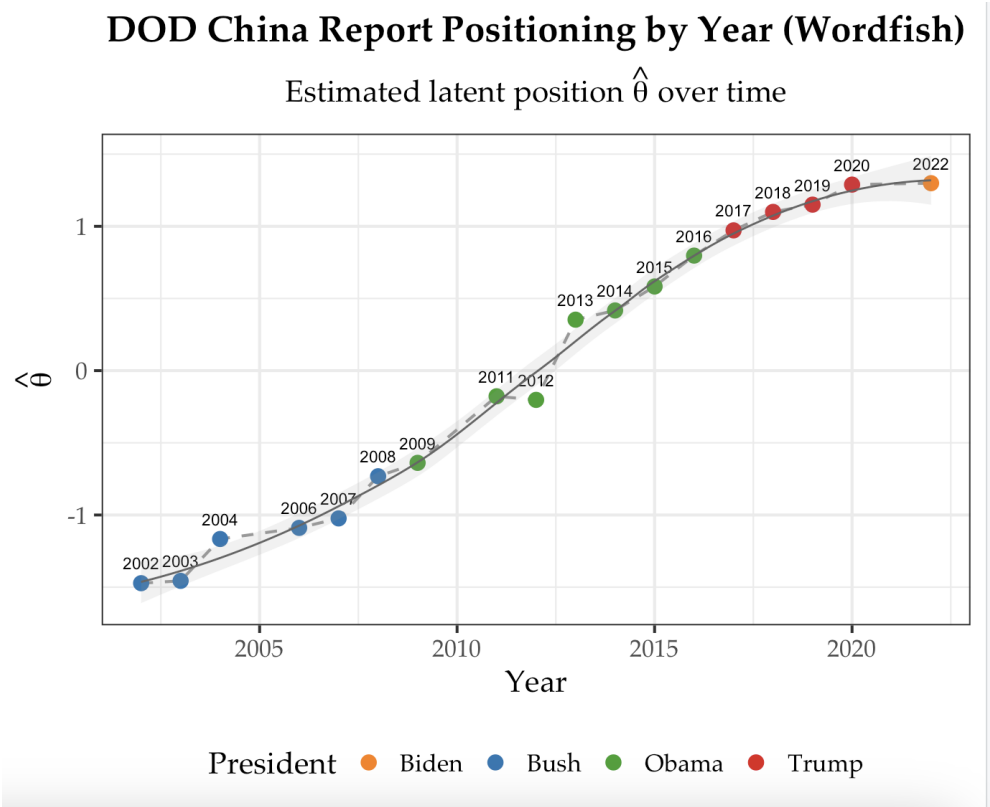


These two plots show the theta score for each document over time, indicating that the language in the documents has changed over time and the reports are linguistically distinct enough. The second image shows the same plot but grouped by president. The

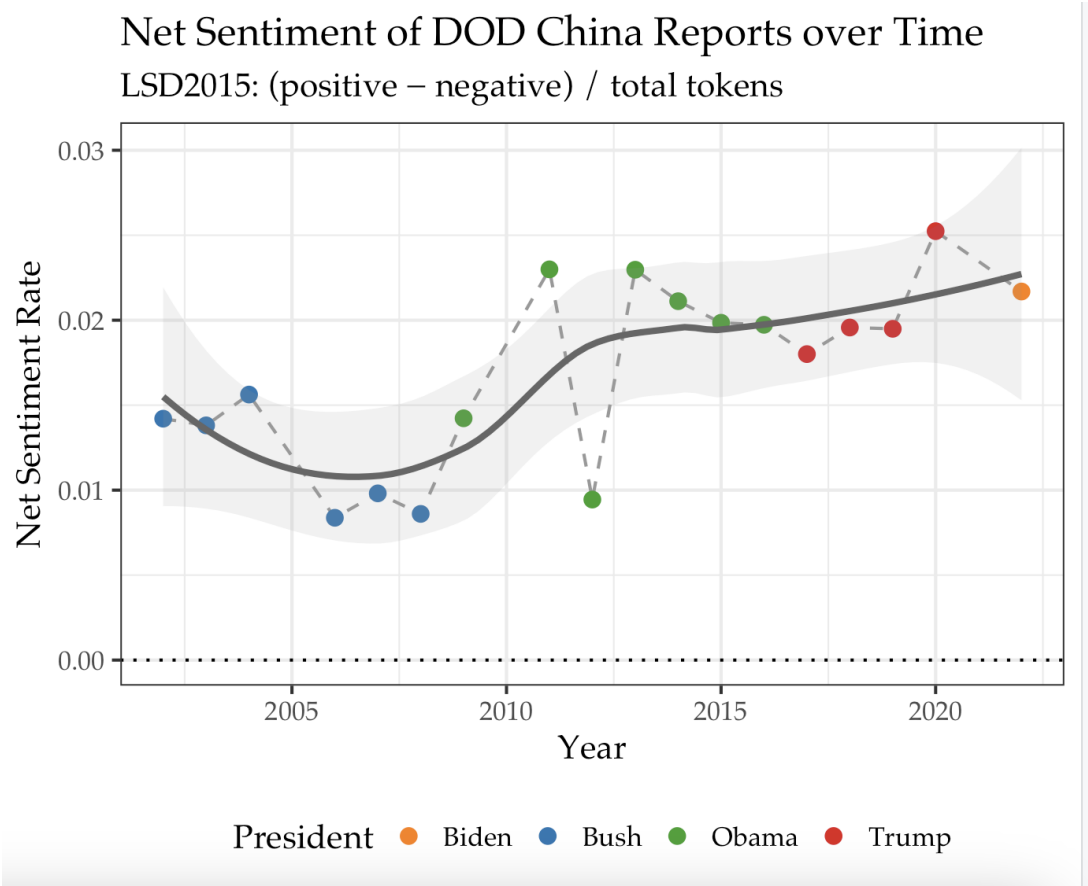
language in the Bush report is far to the left, potentially reflecting post 9/11 related language.

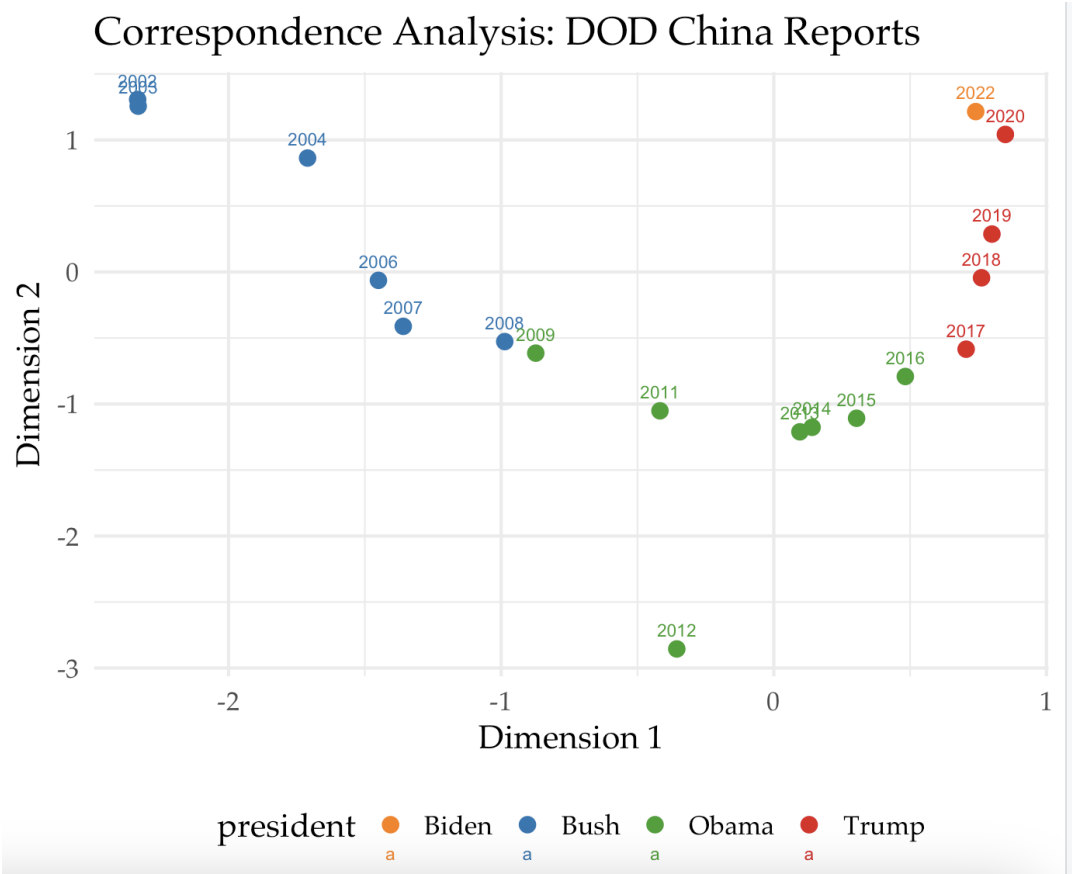


This image shows the beta/psi plot where each dot is a word. The x axis shows how strongly the word is found between early vs late reports, whereas the y axis shows how common the word overall. It makes sense that we see covid-19 far to the right on the plot – it should be completely absent from reports before 2020.



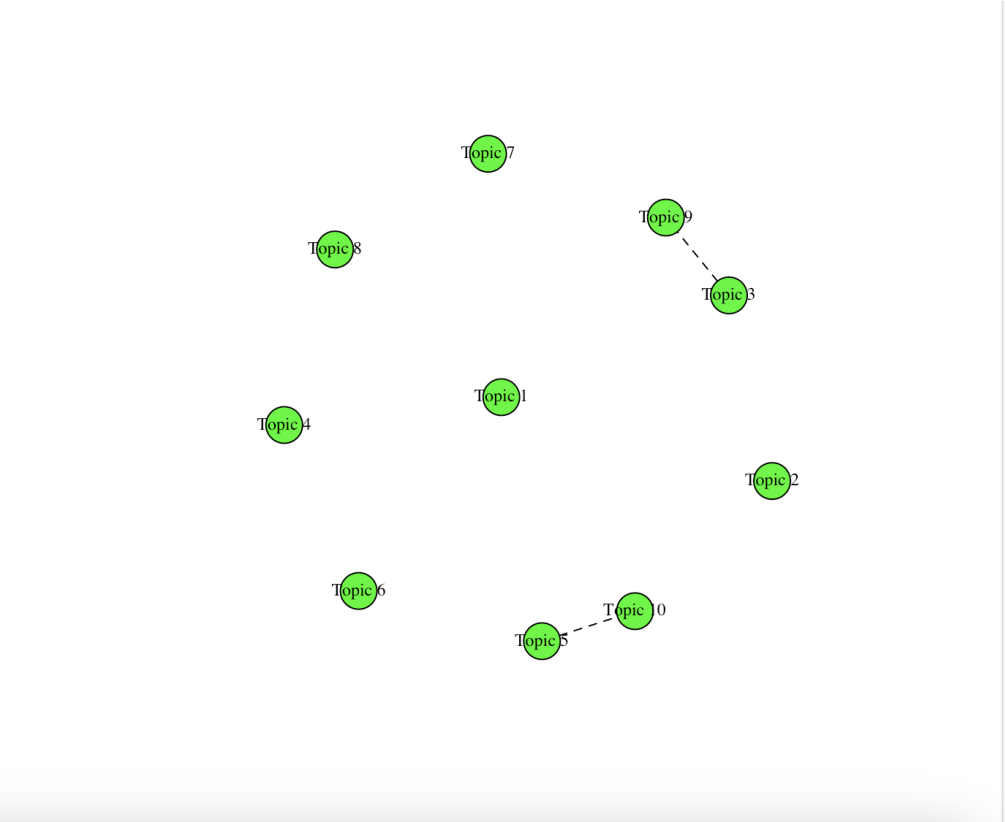
This figure shows a smoothed curve of theta over time. We see a shift in language between the Bush and. Throughout the Obama era, levelling off during the Trump and Biden eras. This suggets the change in DOD language was concentrated from 2010-2017.





Analysis suggested either K=7 or K=10 topics and I went with K = 10.

The topic correlation network figure below shows which topics tend to co-occur in the same document. The analysis suggests that 9 and 3 are connected, as are 5 and 10.



Topic label analysis:

Topic	FREX words	Interpretation
1	admiral, counterpiracy, delegation, xi	**Military diplomacy & leadership**
2	c4isr, weapon, logistics, psywar	**Military capabilities & doctrine**
3	prc's, mcf, rejuvenation, theater, aukus	**Party-state & civil-military fusion**
4	division, counter-terrorism	**Force structure & CT operations**
5	eu, embargo, gdp, estimates	**Economic & arms trade dimensions**
6	a2ad, maritime, humanitarian, navy	**Maritime strategy & A2/AD**
7	rma, fsu, laser, planaf	**Early modernisation & technology**
8	liaoning, counterterrorism, delegation	**Naval development & exchanges**
9	cyber, obor, innovation, takeaways	**Belt & Road, cyber, modernisation**
10	corruption, anti-access, informatization	**Reform, spending & informatisation**

The statistical analysis suggested that only topic 3 was significantly significant when assessing whether topic prevalence differs by president. The topic was much less prevalent under Bush and Obama.

4. How to compare positions (hint: wordfish and scaling methods)

Wordfish provides a method to compare positions of documents on an underlying dimension. Wordfish places documents on a single latent dimension using word frequencies and assumes some words are systematically more common in documents at one end of the scale. The theta score that it produces are directly comparable across documents. This seems to appropriate for the DOD documents, but other, more supervised, methods like wordscores can be used, but require reference documents.